AAV Harvest and Purification

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SUSTech-BIO 2026 iGEM Team

Purpose

Harvest AAV-producing cells and purify cell-associated recombinant AAV using the Takara AAVpro Purification Kit Midi (All Serotypes), Cat. No. 6675. The workflow covers extraction, nuclease treatment, clarification and centrifugal buffer exchange before recovery for titer measurement. [1]

The reference scale is one T225 flask per purification. Cell preparation and the harvest time point belong to AA01; titer measurement and transduction belong to AA03 and AA04. This procedure does not recover AAV from the discarded culture supernatant.

Materials & Equipment

Item	Supplier / identifier	Use / specification
AAV-producing cells	Prepared under AA01	One T225 flask per reference purification [1]
AAVpro Purification Kit Midi (All Serotypes)	Takara; 6675	Four purifications per kit; use the Midi workflow [1]
Extraction and purification reagents	Included in 6675	Extraction Solutions A plus / B; Cryonase Cold-active Nuclease; Precipitators A / B; Suspension Buffer
Millex-HV filter; syringe	Filter included in 6675; compatible syringe	0.45 µm clarification filter
Amicon Ultra-4 centrifugal filter	Included in 6675	100 kDa cutoff; retain the AAV-containing concentrate inside the filter insert
EDTA solution	Laboratory reagent	0.5 M, pH 8.0
Centrifuge and compatible tubes	Laboratory equipment	4°C centrifugation up to 9,000 × g; swinging-bucket rotor at 15°C for Amicon steps
Pipettes, sterile tips and tubes; vortex; 37°C incubator / bath	Laboratory equipment	Liquid handling, mixing and incubation; freezing-compatible tubes if using the pause point
Biological safety cabinet and PPE	Laboratory equipment	Handle recombinant AAV according to the approved local procedure [1]

Before You Start

Use the stated one-T225 reference scale; do not apply Maxi-kit quantities to this Midi kit. Check the sample identity and AA01 harvest time before collecting cells. [1]

Store Cryonase Cold-active Nuclease at -20°C and the other kit components at room temperature. If the Precipitator A stock contains a white precipitate, dissolve it completely at 37°C before dispensing. [1]

Prepare centrifuges for 4°C clarification and 15°C ultrafiltration. Use a swinging-bucket rotor for the Amicon concentration and wash steps. [1]

Distinguish the two filter outputs: the Millex-HV filtrate is retained for the next stage; the Amicon filtrate in the outer tube is discarded while the concentrate inside its filter insert is retained. Prepare a five-wash checklist.

Procedure

A. Harvest the producer cells

1. Add 0.5 M EDTA (pH 8.0) to the culture at 1/80 of the medium volume. Mix and leave at room temperature for 10 min to detach the cells. [1]

2. Collect the cell suspension into a centrifuge tube. Centrifuge at 1,700-2,000 × g for 10 min at 4°C and remove the supernatant. [1]

3. Centrifuge the cell pellet again at 1,700-2,000 × g for 1 min at 4°C, then remove the residual supernatant completely. [1]

B. Extract cell-associated AAV

4. Loosen the pellet by tapping or vortexing until no compact clumps remain. Add 2 mL AAV Extraction Solution A plus and vortex for 15 s to resuspend the cells. [1]

5. Leave at room temperature for 5 min, vortex again for 15 s, then centrifuge at 4,000-9,000 × g for 10 min at 4°C. [1]

6. Transfer the supernatant to a fresh sterile tube without disturbing the pellet. Add AAV Extraction Solution B at 1/10 of the recovered supernatant volume. [1]

Pause point: after adding Extraction Solution B, proceed directly to nuclease treatment or store the extract at -80°C in a suitable tube. Thaw the frozen extract rapidly in a 37°C water bath before continuing. This pause point applies to the extract at this stage. [1]

C. Remove nucleic acids and clarify

7. Add Cryonase Cold-active Nuclease at 1/100 of the AAV extract volume and incubate at 37°C for 1 h. The Chinese manual gives a nominal final activity of 200 U/mL for the supplied enzyme. [1]

8. Add Precipitator A at 1/10 of the sample volume. Vortex for 10 s, incubate at 37°C for 30 min, then vortex again for 10 s. Precipitate formed during this incubation does not prevent the next step. [1]

9. Add Precipitator B at 1/20 of the mixture volume, immediately vortex for 10 s, then centrifuge at 5,000-9,000 × g for 5 min at 4°C. Retain the supernatant. [1]

10. Pass the supernatant through the supplied Millex-HV 0.45 µm filter using a compatible syringe. Collect the filtrate containing AAV. [1]

D. Concentrate and wash in the Amicon filter

11. Load the clarified filtrate into an Amicon Ultra-4, 100 kDa filter insert. Centrifuge at 2,000 × g for 5 min at 15°C in a swinging-bucket rotor. Continue centrifugation if needed until the liquid retained inside the filter insert is below 0.4 mL. [1]

12. Discard the filtrate from the outer collection tube, keeping the filter insert and its concentrate. Add 1 mL Suspension Buffer to the filter insert and mix by pipetting. Centrifuge at 2,000 × g for 5 min at 15°C; continue if needed until the retained volume is below 0.4 mL. [2]

13. Repeat the buffer-addition, mixing and centrifugation cycle four more times, for five washes in total. Count each completed wash. After the last wash, concentrate to the volume needed for subsequent handling and record the volume recovered. [1, 2]

Version note: the English manual specifies <0.4 mL after each wash; the Chinese manual gives <1.5 mL at that step. This protocol consistently uses the English wash endpoint. Both versions specify five washes. [1, 2]

If a small retained volume is difficult to see low inside the filter cup, do not interpret this alone as sample loss. Continue the prescribed Suspension Buffer washes, keeping the filter insert and its contents; discard only the filtrate in the outer collection tube.

E. Recover the purified AAV

14. Discard the final Amicon filtrate. Resuspend the concentrate inside the filter cup by pipetting or vortexing for 30 s, then transfer it to a fresh tube. [1]

15. Label the recovered suspension with sample identity and purification date, record its volume, and take an appropriate aliquot for AA03 titer measurement. Use the established downstream handling plan for the remaining material.

p.s.

Small volumes may be difficult to see during ultrafiltration. If this is only a visibility issue, continue the scheduled washes rather than adding extra centrifugation simply to make the sample visible. Keep the AAV-containing concentrate inside the filter insert.

Remove residual medium thoroughly before adding Extraction Solution A plus, and disperse the pellet well. Residual medium and persistent cell clumps can impair purification. [1]

A pink color after adding Extraction Solution B, or precipitate during the Precipitator treatment, can occur during the normal workflow. Follow the stated clarification step rather than rejecting a sample on appearance alone. [1]

The documented -80°C pause point is after addition of Extraction Solution B. The -20°C storage condition in the kit list refers to Cryonase, not to a validated storage condition for the recovered AAV preparation. [1]

Expected Outcome

Recover an AAV-containing concentrate in Suspension Buffer. Record the recovered volume and measure vector-genome titer under AA03; volume or visual appearance alone does not establish yield or purity.

If assessed, capsid-protein analysis and a separate transduction assay provide information about purity and functional activity. A vector-genome titer alone does not establish functional potency. No fixed yield is assumed for a new preparation.

Troubleshooting

Observation	Check first	Action
Cell clumps persist	Pellet loosening and residual culture medium	Disperse the pellet and resuspend until no clumps remain before continuing extraction. [1]
White precipitate in Precipitator A stock	Storage temperature; distinguish stock precipitate from reaction precipitate	Dissolve stock precipitate fully at 37°C before dispensing. Reaction precipitate is handled by the stated workflow. [1]
Retained volume remains above the endpoint	Rotor type, × g setting, 15°C setting and retained volume	Continue centrifugation under the stated conditions and recheck volume. For this protocol, use <0.4 mL after initial concentration and each wash. [1, 2]
Low recovered amount or titer	AA01 input, medium removal, cell dispersion, transfers and retained compartment	Check AA03 assay validity and handling records first. Recover material from the filter insert; do not substitute the outer-tube filtrate for the concentrate.

Reference

[1] Takara Bio. AAVpro Purification Kit Midi (All Serotypes), Code No. 6675. Chinese manual v202004Da, printed pp. 3-5 (PDF pp. 5-7). User-provided local copy 6675.pdf; matched official online copy checked 2026-10-03. Open source

[2] Takara Bio. AAVpro Purification Kit Midi (All Serotypes), Cat. No. 6675 / 6675S. English manual v202001Da, p. 8, section VI-2, especially the wash-volume endpoint. Accessed 2026-10-03. Open source